Marcelo Machuca

AI Solutions Developer

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SUMMARY

Applied AI developer: tool-using agents, RAG with citations and cost measurement, in Python, TypeScript and Rust. I design the architecture and direct coding agents (Claude Code, Codex, Gemini) on the implementation; nothing merges without tests, evals and diff review. I write the hypothesis down before measuring and publish the result, negative ones included. Open to AI or Python developer roles, fully remote.

PROJECTS

Independent work, Apr 2026 – present. Every number comes from a test, an eval or a commit.

PUBLIC

Farol Desk · IT service-desk agent (formerly Nexo Assist) — github.com/skarL007/nexo-assist

`TypeScript · Next.js 16 · Mastra · DeepSeek-V4-Flash · Drizzle · Vitest · Playwright · demo: nexo-assist-puce.vercel.app`

- Agent that triages the service desk: checks and opens tickets only after user confirmation, searches the help center and hands the chat to an analyst; every answer shows its steps, tools, tokens and latency.
- Two-stage search (hybrid + relevance judge, Corrective RAG pattern): on 81 test questions (30 Oct), 78 got the right article and none got a wrong one; without the judge it was 67 right and 10 wrong.
- Measured quality: 573 unit tests and 48 E2E (11 Nov); evals 35/36 with the real model and red team 21/21 (3 Nov); simulated employee pass^3 from 0.50 to 0.80 after the fixes (30 Oct round, not re-run with the current model).
- CI on CircleCI, an image on Docker Hub and continuous deployment on Komodo; public demo with a usage cap.

Lodemap · architecture map that fails the build when it falls behind — github.com/skarL007/lodemap

`Python · Structurizr DSL · JavaScript · Claude Code skill · MIT`

- Keeps a C4 map in the repository and fails the build when the code grows and the map does not; CI green on Linux, macOS and Windows. Demo on axios: 72 of 72 modules on the map, checked by a structural test.
- Pre-registered experiment, 96 runs, negative result published: the map did not make the agent cheaper (median ratio 0.99) and the answers were not worse.

LumenAI SDK · per-tenant LLM cost on OpenTelemetry — github.com/skarL007/lumen-ai-sdk

`Python · OpenTelemetry · Redis · Celery`

- SDK that turns LLM call spans into per-tenant cost, exporting metadata only: no prompt or response leaves the application.
- Checked against a real bill: across 6 calls through OpenRouter, the computed cost matched the charged cost on every call, cache included. On PyPI (lumen-ai-sdk 0.1.6), 180+ tests and budget alerts.

VoiceLaunch TTS · voice accessibility app — github.com/skarL007/sound_voice

`TypeScript · Electron · React · Python · FastAPI`

- Open-source desktop app that turns typed text into speech on a virtual microphone for Discord and games, switching to a local voice (Kokoro, Piper) when offline. 150+ tests (Vitest, Playwright) and a Windows installer.

PRIVATE (I WALK THROUGH THE CODE IN AN INTERVIEW)

Lodeward · pre-execution firewall for coding agents (formerly SynapseEngine) — private

`Rust · MCP · Claude Code hooks`

- MCP server and hooks that inspect every shell command the agent proposes before it runs. 988 commits on main (13 Sep–10 Nov 2026); failing test rounds are logged, not hidden.

Sublinhe · RAG with citations for health students — private

`Python · FastAPI · BGE-M3 · FAISS · BM25 · reranker · React`

- Turns lecture material into flashcards, questions and oral practice, always citing the page, and abstains when the source does not support an answer. RRF fusion and LLM-judge evals; 1,140 commits on main, MVP closed on 5 Oct 2026.

Aulô · voice study platform (in progress) — private · in progress

`Python · JavaScript`

- Students upload their own material and a virtual teacher answers only from it, showing the page it cites. Not launched yet.

SKILLS

Languages	Python, TypeScript, JavaScript, Rust, SQL
Applied AI	LLMs, tool-using agents (Mastra, LangGraph), RAG (embeddings, FAISS, BM25, reranking), MCP
Evaluation	evals, red teaming, user simulation (pass^k), pre-registered experiments, Pytest, Vitest, Playwright
Backend and data	FastAPI, Next.js, Node.js, PostgreSQL, SQLite/libSQL, Redis, OpenTelemetry
Infra	Docker, CircleCI, GitHub Actions, Vercel, Git

EDUCATION AND LANGUAGES

Medicine	Amazonas State University (UEA), Brazil · left after the 8th of 12 semesters
Languages	Portuguese (native) · English (intermediate)